Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 1 (Generated practice paper - Set 35)

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
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6	
7	
8	
9	
10	
11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 . 1 Which of the following is a feature of a river that affects its usefulness as a water source? Tick **one** box.

[1 mark]

A Its total annual discharge ☐

B The albedo of its surface ☐

C Its salinity ☐

D The age of its bedrock ☐

0 1 . 2 Explain why rainwater collection is increasingly important in cities such as those in India.

[2 marks]

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0 1 . 3 Explain why an estuary barrage has fewer land-use conflicts than a reservoir on land but still causes environmental problems.

[2 marks]

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0 2 . 1 Which of the following best describes the purpose of Critical Group Monitoring? Tick one box.

[1 mark]

A To detect health damage in the population most affected by a pollutant ☐

B To predict the route a pollutant will take through the environment ☐

C To assess exposure of the people most at risk so that everyone else can be assumed to be sa ☐

D To measure the concentration of a pollutant at the point of release ☐

0 2 . 2 Explain why CPA can reduce the cost of an environmental monitoring programme.

[2 marks]

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0 2 . 3 A new chemical has properties similar to PCBs (persistent, liposoluble chlorinated organic compounds). Explain how knowledge of PCBs helps decisions about the new chemical.

[2 marks]

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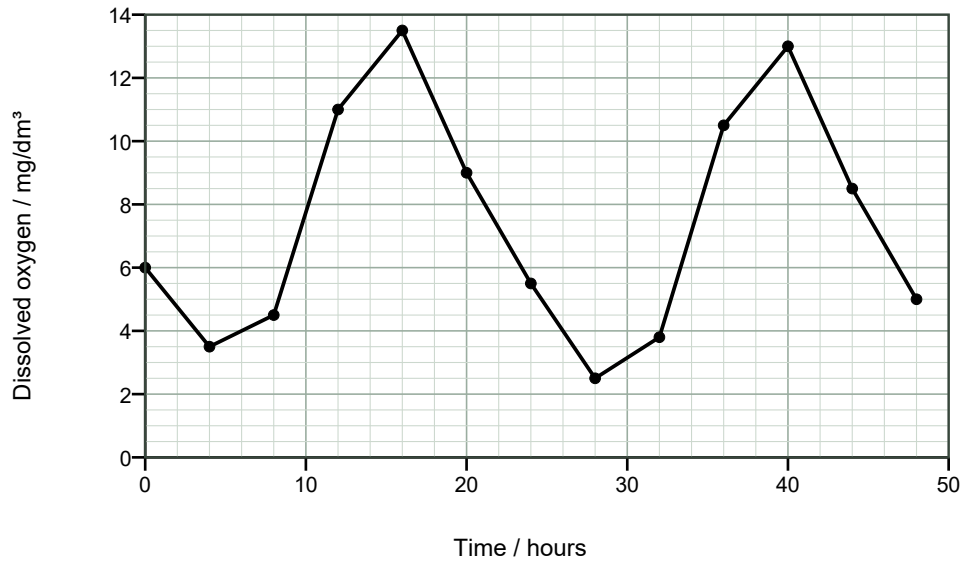
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0 3 Figure 1 shows the dissolved oxygen concentration in a shallow lowland lake over 48 hours in late summer. The lake receives runoff from arable farmland and effluent from a sewage works.

Figure 1



0 3 . 1 Describe the pattern shown in Figure 1.

[2 marks]

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0 3 . 2 Explain the causes of the pattern shown in Figure 1.

[4 marks]

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0 3 . 3 Explain why fish kills in eutrophic lakes usually happen at night or at the end of summer.

[2 marks]

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0 3 . 4 Give **two** other harmful effects of algal blooms.

[2 marks]

1 .....

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2 .....

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10

0 4 . 1 Describe the main features of the Montreal Protocol (1987).

[3 marks]

0 4 . 2 For each of the following CFC uses, give the alternative process or material now used and one disadvantage of the alternative.

[4 marks]

Aerosol propellant

Refrigerant

0 4 . 3 Describe how waste CFCs from old refrigerators are disposed of safely.

[2 marks]

0 4 . 4 State **one** reason why the ozone layer is not expected to recover fully for many decades.

[1 mark]

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0 5 . 1 Explain how each of the following human activities changes the amount of carbon in the atmosphere.

[6 marks]

Ploughing farmland .....

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Draining marshland / peat bogs .....

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Toxic pollution of the sea .....

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0 5 . 2 Give **three** anaerobic environments created by human activities that release methane.

[3 marks]

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0 5 . 3 Explain how rising atmospheric CO_2 causes ocean acidification.

[1 mark]

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0 6 A 1960s office block is being refurbished. Table 1 shows the measured heat loss through its fabric before the work.

Table 1

Part of the building	Heat loss / kW
Walls (uninsulated)	42
Roof	26
Single-glazed windows	31
Floor	9
Draughts and ventilation	22

0 6 . 1 Calculate the percentage of the total heat loss that passes through the windows.

[2 marks]

Answer ..... %

0 6 . 2 Explain how each of the following reduces heat loss.

[3 marks]

Cavity wall insulation .....

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Low-emissivity double glazing .....

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A heat exchanger on the ventilation system .....

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0 6 . 3 Explain why there is an optimum thickness of loft insulation rather than simply using as much as possible.

[2 marks]

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0 6 . 4 The refurbishment also includes a combined heat and power (CHP) unit. Explain why a CHP unit has a lower **electrical** efficiency than a large power station but a much higher **overall** efficiency.

[3 marks]

0 6 . 5 Explain **five** other ways the refurbished building could be designed or managed to use less energy.

[5 marks]

15

0 7 . 1 Which chemical form of mercury is most easily absorbed and biomagnified in food chains? Tick **one** box.

[1 mark]

A Elemental (liquid) mercury ☐

B Inorganic mercury compounds ☐

C Organic (methyl) mercury ☐

D Mercury vapour ☐

0 7 . 2 Explain why anaerobic sediments increased the hazard of the inorganic mercury discharged at Minamata.

[2 marks]

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0 7 . 3 Explain why heavy metal wastes are stored under alkaline conditions.

[2 marks]

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0 8 A student measured the wind speed at 10 sites inside a woodland and at 10 sites in a nearby clearing.

Table 1

Location	Mean wind speed / m s^{-1}	Standard deviation / m s^{-1}
Woodland	1.2	0.4
Clearing	3.5	1.1

0 8 . 1 Name the statistical test the student should use to find out whether the difference between the means is significant. Give **two** reasons for your choice.

[3 marks]

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0 8 . 2 Explain what the student should do if it is not possible to show that the data are normally distributed.

[1 mark]

0 8 . 3 The test gave p = 0.02. Explain what this means.

[2 marks]

0 8 . 4 Explain how the standard deviations in Table 1 support the conclusion that the means are different.

[2 marks]

0 8 . 5 Explain why a statistically significant result does not prove that trees cause the lower wind speed.

[2 marks]

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0 9 . 1 Explain **three** features of fossil fuels that made them ideal for the development of industrial society.

[3 marks]

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0 9 . 2 75 t of aviation fuel carries a Boeing 747 and its 400 passengers from London to New York. Calculate the mass of fuel used per passenger, in kg.

[2 marks]

Answer ..... kg

0 9 . 3 Explain why the total quantity of fossil fuels in existence is much greater than the quantity counted as **reserves**.

[3 marks]

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0 9 . 4 Cheap North Sea gas contributed to the closure of the UK's deep coal mines, although large coal deposits remained. Explain why those mines cannot now be reopened.

[2 marks]

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0 9 . 5 Explain **four** environmental impacts of the **extraction** of fossil fuels, as distinct from burning them.

[5 marks]

15

1 0 . 1 Using named examples, explain how the sources of noise pollution can be controlled by changes to design, operation and location.

[9 marks]

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1 0 . 2 State what is meant by the **Leq 57 dB** contour used around UK airports.

[1 mark]

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Write an essay on one of the following topics.

11 . 1 Explain how feedback mechanisms and tipping points influence the rate of global climate change and discuss the implications for the strategies needed to control it.

[25 marks]

OR

11 . 2 Discuss how the atmosphere supports life on Earth and evaluate the extent to which human activities threaten these life-support functions.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS